

# Marathon Performance Analysis: Impact of Different Weather Conditions on Runners

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## Introduction

Marathon running is a popular sport that attracts millions of participants worldwide. The completion time of a marathon is influenced by various factors, including weather conditions. A 42-km marathon footrace is among the most physiologically demanding endurance events, with competitive runners typically maintaining a pace corresponding to 70-90% of their maximal aerobic capacity for more than 2 hours. At maximal mechanical efficiency, more than 80% of the energy required is transferred as heat to the body core, producing approximately 1400 kcal/hour that must be dissipated to prevent dangerous rises in body temperature.

While numerous studies have documented performance degradation in hot environments, research has largely focused on elite male runners, leaving the implications for slower competitors and women runners largely speculative. Additionally, there has been limited research examining performance when tasks elicit only modest elevations in core body temperature ( $<38.5^{\circ}\text{C}$ ). In this report, we analyze the impact of different weather conditions on marathon performance across the lifespan in both men and women using data from five major marathons: Boston, NYC, Chicago, Twin Cities, and Grandmas. We aim to examine effects of increasing age on marathon performance in men and women and the impact of environmental conditions on marathon performance, and whether the impact differs across age and gender. Additionally, we want to find out which factor has the most significant impact on marathon performance.

## Data Description

Two datasets are used in this analysis: Marathon Data and Course Record Data. Those two datasets were combined to form a comprehensive dataset for analysis.

### Marathon Data

The Marathon Data contains information about the top completion record percentage grouped by age and sex of runners in five major marathons: Boston, NYC, Chicago, Twin Cities, and Grandmas. The dataset includes the following columns:

| Parameter                                | Coding details                                                                                                                                                                              |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Race</b>                              | 0 = Boston Marathon , 1 = Chicago Marathon , 2 = New York City Marathon , 3 = Twin Cities Marathon (Minneapolis, MN) , 4 = Grandma's Marathon (Duluth, MN)                                  |
| <b>Year</b>                              | Year of the marathon                                                                                                                                                                        |
| <b>Sex/Gender</b>                        | 0 = Identified as Female , 1 = Identified as Male                                                                                                                                           |
| <b>Flag</b>                              | White = WBGT $<10^{\circ}\text{C}$ , Green = WBGT $10-18^{\circ}\text{C}$ , Yellow = WBGT $18-23^{\circ}\text{C}$ , Red = WBGT $23-28^{\circ}\text{C}$ , Black = WBGT $>28^{\circ}\text{C}$ |
| <b>% CR</b>                              | Percent off current course record for gender                                                                                                                                                |
| <b>Td, <math>^{\circ}\text{C}</math></b> | Dry bulb temperature in Celsius                                                                                                                                                             |
| <b>Tw, <math>^{\circ}\text{C}</math></b> | Wet bulb temperature in Celsius                                                                                                                                                             |

| Parameter                 | Coding details                             |
|---------------------------|--------------------------------------------|
| <b>%rh</b>                | Percent relative humidity                  |
| <b>Tg, °C</b>             | Black globe temperature in Celsius         |
| <b>SR W/m<sup>2</sup></b> | Solar radiation in Watts per meter squared |
| <b>DP</b>                 | Dew Point in Celsius                       |
| <b>Wind</b>               | Wind speed in Km/hr                        |
| <b>WBGT</b>               | Wet Bulb Globe Temperature                 |

Note that the variable **WBGT** is actually a composite index that combines the effects of **Td**, **Tw**, and **Tg**, which means we can ignore the three variables when analyzing the data.

$$WBGT = 0.7 \times Tw + 0.2 \times Tg + 0.1 \times Td$$

This data contains of 11073 observations and 12 variables. The data documented the 5 major races form year 1993 to 2016, and each observation represents the average performance of a specific age in a particular race during a specific year. It should be noted that in this dataset, only one set of weather data is recorded for each race.

### Course Record Data

This dataset contains the course record for each race grouped by sex documented in the Marathon Data. The dataset includes the following columns:

| Parameter         | Coding details                                                                                                                                               |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Race</b>       | B = Boston Marathon , C = Chicago Marathon , NY = New York City Marathon , TC = Twin Cities Marathon (Minneapolis, MN) , D = Grandma's Marathon (Duluth, MN) |
| <b>Sex/Gender</b> | 0 = Identified as Female , 1 = Identified as Male                                                                                                            |
| <b>Year</b>       | Year of the marathon                                                                                                                                         |
| <b>CR</b>         | Current course record for gender                                                                                                                             |

Different from the Marathon Data, the **CR** in this dataset is in hours, which means we can calculate the completion time of each runner by multiplying the CR percentage in the Marathon Data with the CR in this dataset.

## Data Preprocessing

### Data Manipulation

For clarity, we firstly rename the **Race** column in the Marathon Data as well as the Course Record Data to the full name of the marathon. And since we are using two datasets, we need to merge them based on the **Race**, **Year**, and **Sex** columns. After merging, we now have a united dataset that contains 15 variables and 11073 observations in total.

And then, condiering of the nature of the variables and the analysis we are going to perform, we convert the **Year**, **Race**, **Sex**, and **Flag** columns to factors. We also created a new column **Age\_group** by grouping the **Age** column into 10-year intervals. This step is essential trying to factorize the **Age** column.

Most importantly, we would like to convert the **CR** in the Course Record Data to seconds. The process is simple, we first convert the **CR** to a period object and then convert it to seconds. After that, we can calculate the completion time of each runner by multiplying the CR percentage in the Marathon Data with the CR in this dataset. The calculation formula is as follows:

$$CR_{adjusted} = (1 + CR_{PERCENTAGE} \times 0.01) \times CR$$

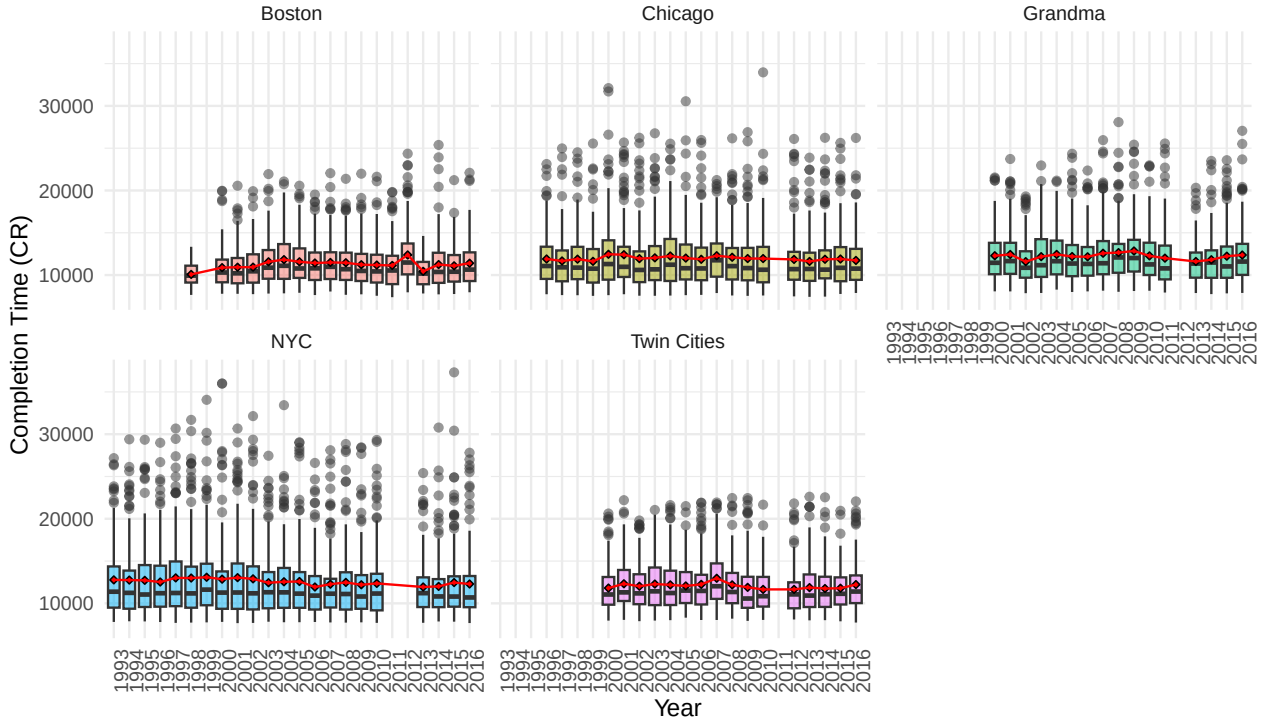
## Data Quality Check

Before we start the analysis, we need to check the data quality. We first check for missing values and patterns in the dataset. The missing values are visualized using the `vis_miss` function from the `naniar` package. Results shows that there are missing values in the weather related columns, and this missingness seems to have a very clear pattern. Based on this pattern, we can assume boldly that the missing values are not missing at random, but are related to specific races.

In order to verify our assumption, we calculate the missing percentage of weather data in each marathon by year. Results shows that the missing percentage of most of the races are 0, however, for races held in 2011 (Chicago, NYC, and Twin Cities) as well as the one held in 2012 (Grandmas), the missing percentage is 100%. This confirms our assumption that the missing values are related to specific races. It would be a wise choice to remove all of those races from our dataset. To be noted that the NA value in the table means there are no races held in that year.

And we also want to check the performance distribution of each race by year, since although the weather condition may vary from year to year, the performance of runners may not change significantly within the same track. As we can see in Figure 2, despite some disturbances, the performance of runners tend to stay stable over the years. The difference in performance might be due to the different weather conditions in different years. Overall, the stability of the performance indicates that the dataset is reliable for analysis.

Figure 2: CR Distribution by Year and Race



Another factor might affect the result of the analysis is **age**. In a balanced dataset, we would expect the number of participants in each age group to be roughly the same. To check this, we count the number of participants in each age group for each races. The table shows a pattern that for each race, participants between age 20 to 79 are the most common and also balanced, while the number of participants in the 10-19, 80-89 and 90-99 age groups are relatively small. This suggests that the inference results obtained from the 10-19, 80-89 and 90-99 age groups may not be as reliable as those obtained from the 20-79 age groups.

Moreover, the number of participants in the 90-99 age group is extremely small, so we decide to merge it with the 80-89 age group to become the 80-99 age group.

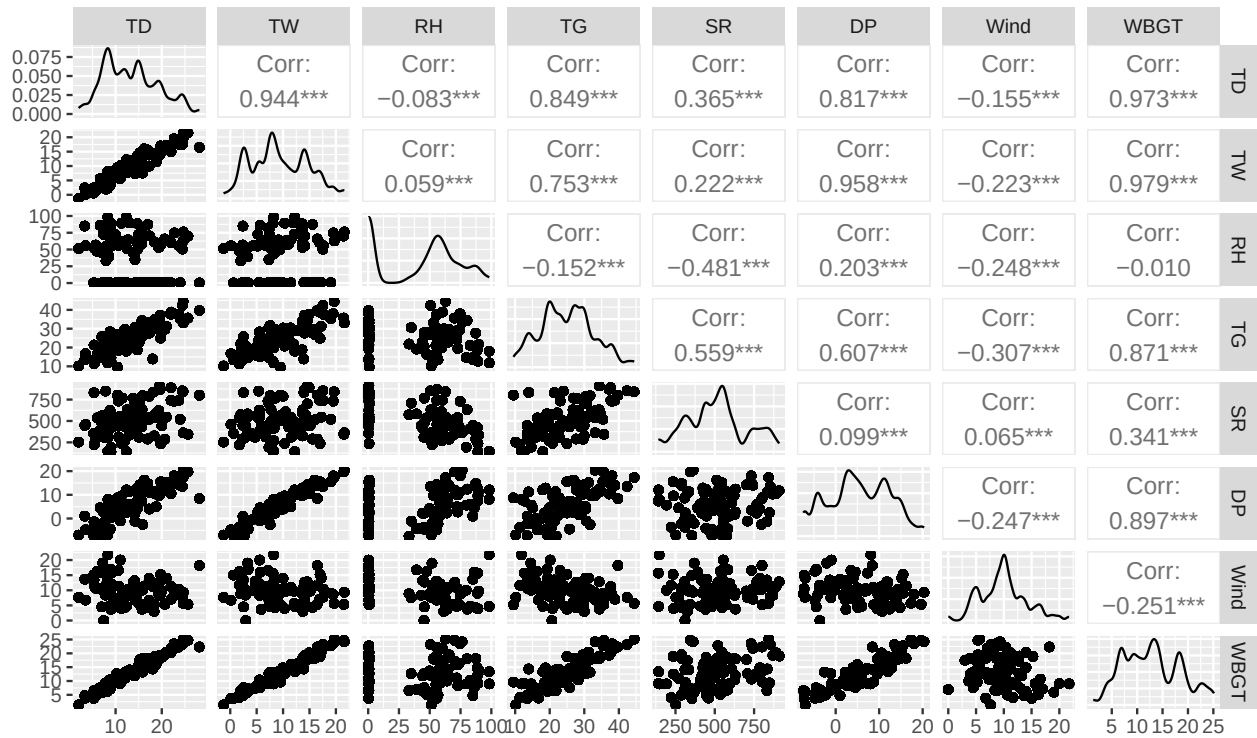
Similarly, we would hope that participants' gender is balanced in the dataset. From the table below, we can tell the Female:Male portion is roughly equal to 1 in all of the races, which is a good feature for our analysis.

Table 3: **Number of Participants by Age Group, Sex and Race**

| <b>Age Group</b> | <b>Boston, N =<br/>2,088</b> | <b>Chicago, N =<br/>2,427</b> | <b>Grandma, N =<br/>1,884</b> | <b>NYC, N =<br/>2,799</b> | <b>Twin Cities, N =<br/>1,875</b> |
|------------------|------------------------------|-------------------------------|-------------------------------|---------------------------|-----------------------------------|
| Age_group        |                              |                               |                               |                           |                                   |
| 0-9              | 0 (0%)                       | 0 (0%)                        | 0 (0%)                        | 0 (0%)                    | 0 (0%)                            |
| 10-19            | 67 (3.2%)                    | 171 (7.0%)                    | 167 (8.9%)                    | 88 (3.1%)                 | 140 (7.5%)                        |
| 20-29            | 360 (17%)                    | 400 (16%)                     | 320 (17%)                     | 440 (16%)                 | 319 (17%)                         |
| 30-39            | 360 (17%)                    | 400 (16%)                     | 320 (17%)                     | 440 (16%)                 | 320 (17%)                         |
| 40-49            | 360 (17%)                    | 400 (16%)                     | 320 (17%)                     | 440 (16%)                 | 320 (17%)                         |
| 50-59            | 359 (17%)                    | 400 (16%)                     | 320 (17%)                     | 440 (16%)                 | 319 (17%)                         |
| 60-69            | 337 (16%)                    | 391 (16%)                     | 286 (15%)                     | 438 (16%)                 | 294 (16%)                         |
| 70-79            | 215 (10%)                    | 237 (9.8%)                    | 134 (7.1%)                    | 378 (14%)                 | 146 (7.8%)                        |
| 80-89            | 30 (1.4%)                    | 28 (1.2%)                     | 17 (0.9%)                     | 130 (4.6%)                | 17 (0.9%)                         |
| 90-99            | 0 (0%)                       | 0 (0%)                        | 0 (0%)                        | 5 (0.2%)                  | 0 (0%)                            |
| Sex              |                              |                               |                               |                           |                                   |
| Female           | 984 (47%)                    | 1,150 (47%)                   | 880 (47%)                     | 1,337 (48%)               | 867 (46%)                         |
| Male             | 1,104 (53%)                  | 1,277 (53%)                   | 1,004 (53%)                   | 1,462 (52%)               | 1,008 (54%)                       |

We also want to check the correlation between the weather variables. Figure 3 shows that DP is highly correlated with WBGT, which means we can ignore the DP variable when analyzing the data. And since the WBGT is a composite index that combines the effects of Td, Tw, and Tg, all of those variables are highly correlated with WBGT, meaning that they can be ignored as well.

Figure 3: Correlation Plot of Weather Variables

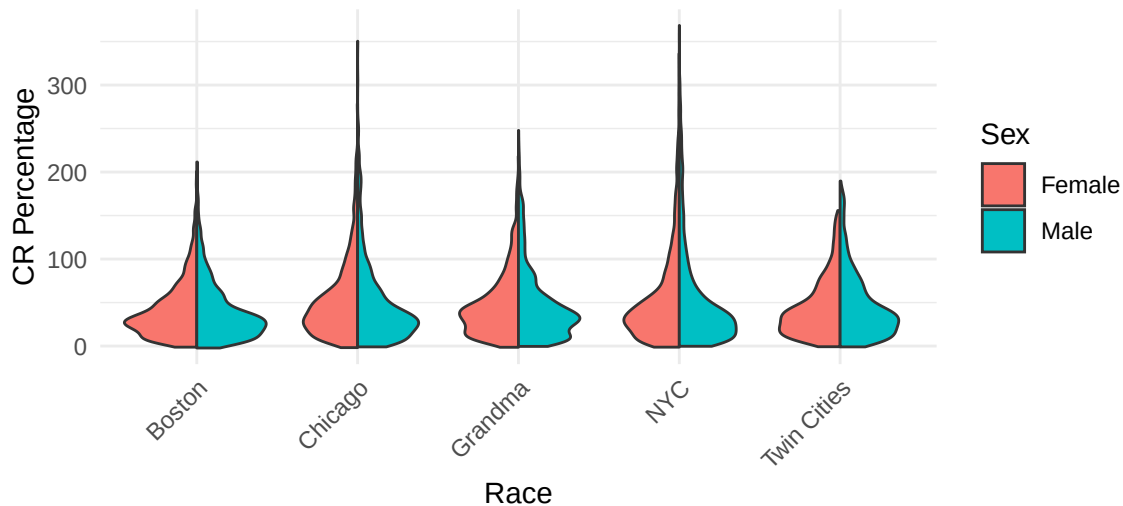


## Data Analysis

### Inner Factors Analysis

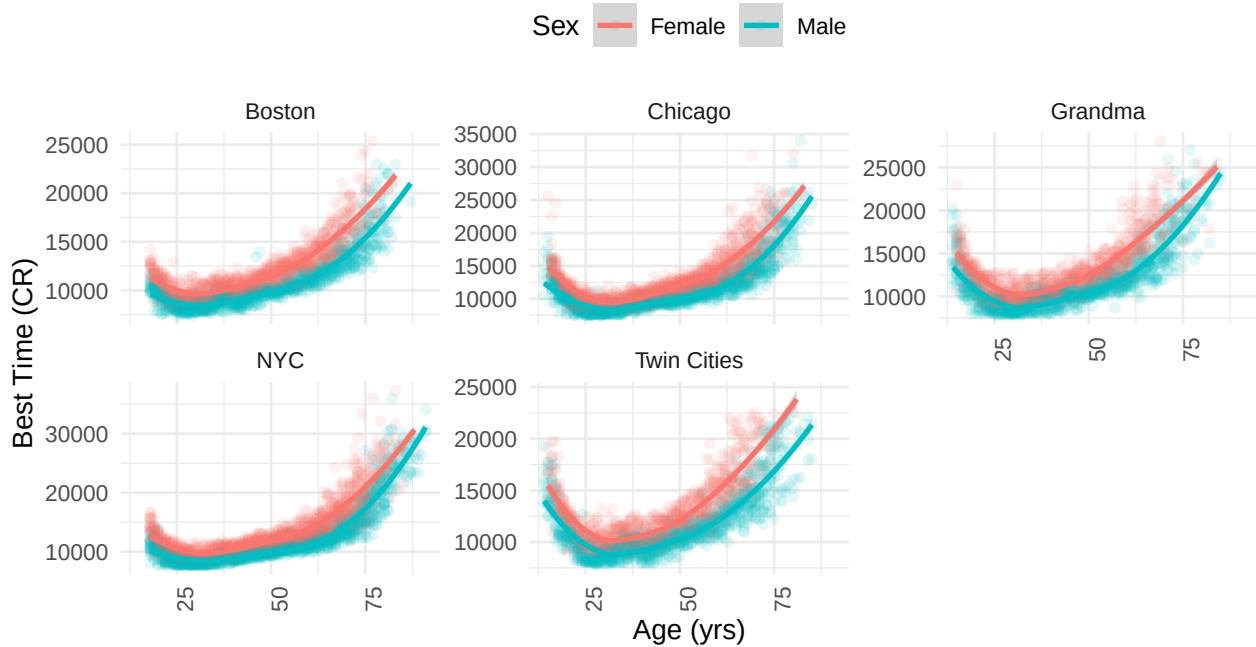
A very instinctive thought is that the completion time of a marathon is influenced by gender. To verify this, we first plot the distribution of completion time percentage among Female and Male, and find that in each races, males tend to have a lower completion time. And we can also observe a distribution pattern in Figure 4 that the completion time for both male and female runners is heavily skewed, which means that most of the runners have a completion time close to the course record. This suggests that our runners are generally well-trained and have a good performance.

Figure 4: Distribution of Completion Time Percentage



Next, we want to explore the effect of age on marathon performance. From Figure 5, we can not only observe the fact that female runners tend to run slower than male runners, but we can also observe that in each case, no matter male or female, the completion time decreases first and then increases with age. And people around their 30s tend to have the best performance. This suggests that the completion time of a marathon is influenced by age, and the effect of age on completion time is not linear.

Figure 5: Effect of Age on Marathon Performance by Race



## Weather (Outer) Factors Analysis

Both of **Age** and **Sex** can be considered as inner factors that affect the completion time of a marathon. We also care about the outer factors such as weather condition. Multiple weather variables are recorded in the dataset, including **RH**, **SR**, **DP**, **Wind**, and **WBGT**. In the following section, we will analyze the effect of these weather variables on marathon performance, and also explore the interaction between weather variables and inner factors.

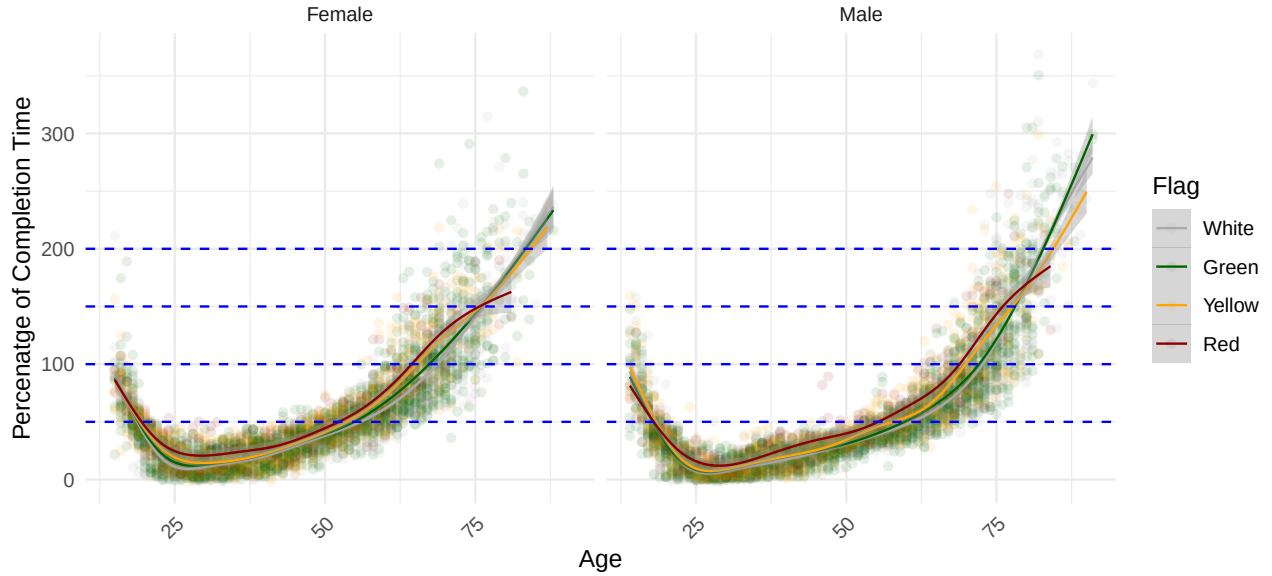
We first want to focus on the effect of **WBGT** on marathon performance. **WBGT** is a composite index that combines the effects of **Td**, **Tw**, and **Tg**. Conceptually, the **WBGT** is an index used to estimate the effect of temperature, humidity, wind speed, and solar radiation on humans, typically to assess heat stress during physical activities in outdoor environments. That suggests the higher the **WBGT**, the more difficult for runners to finish the marathon.

In order to prove our assumption, we plot the effect of **WBGT** on completion time by sex. However, instead of using **WBGT** directly, we use the **Flag** column to represent the **WBGT** level. The **Flag** column is a categorical variable that represents the **WBGT** level, with **White** representing the lowest **WBGT** level and **Black** representing the highest **WBGT** level. In Figure 6, each of the plot represents an observation, and the color of the plot represents the **Flag** level. To better read the result, we also add linear regression line to each plot.

By interpreting Figure 6, we can tell that in general cases, the worse the **WBGT** level, the higher the completion time. In detail, we found that the **WBGT** level do have different effect for different age groups as well as gender. To be note that, the wider the space between two regression lines, the more significant the effect of **WBGT** level on completion time. In that case, we can tell the **WBGT** level has a more significant effect on male than female. And looking at the difference among age groups, we found that the **WBGT** has more effect on elder runners than younger runners. Additionally, we see intersection between the regression lines around age 75, we think

that might be due to the small number of participants in the 70-79 age group, causing the result to be less reliable, the larger standard error of the regression line showing in Figure 6 also suggests this assumption.

Figure 6: WGBT Effect on Completion Time by Age and Sex

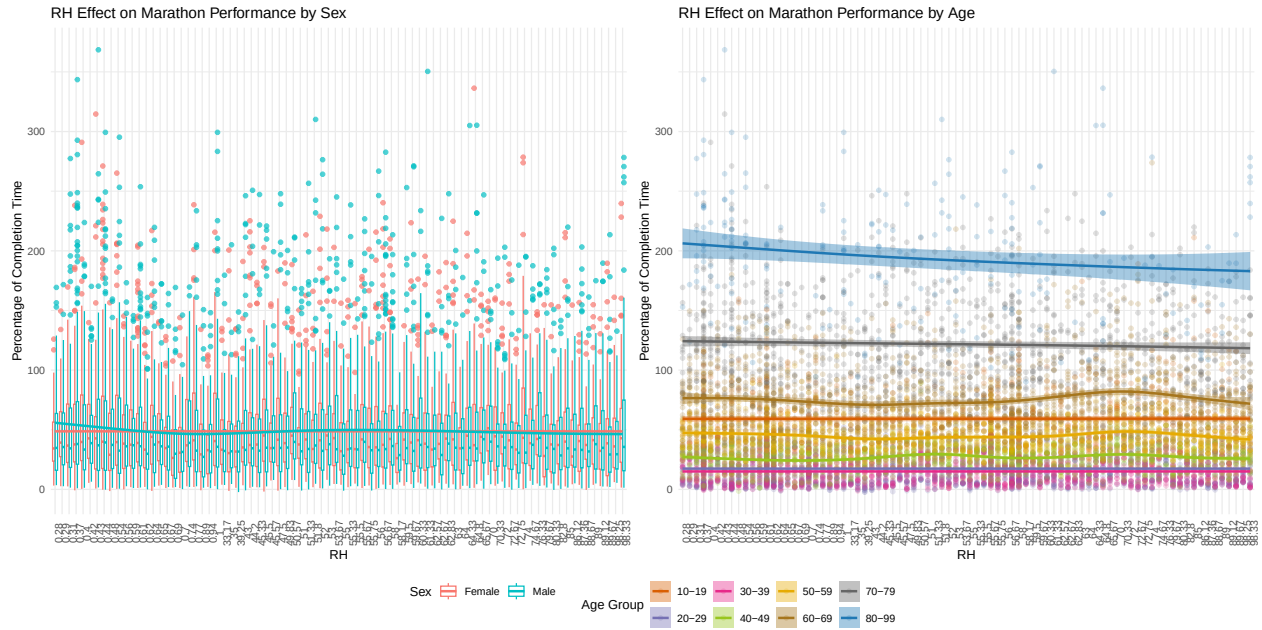


Next, let's check the rest of the weather variables. **RH** is the relative humidity, which is the ratio of the amount of water vapor present in the air to the maximum amount that the air can hold at that temperature. **SR** is the solar radiation, which is the power received per unit area from the sun in the form of electromagnetic radiation in the wavelength range of the solar spectrum. **Wind** is the wind speed, which is the rate at which air flows past a point on the earth's surface in a unit of time.

Since all of those variables are continuous, we set two plots for each of them. The first plot is the boxplot used to show the effect on gender. The weather variable is set as the x-axis, and the completion time percentage is set as the y-axis. The color of the box represents different gender. The second plot is the scatter plot used to show the effect on age. The setting of x-axis and y-axis is same as the first plot. The color of the point represents different age groups. In both of the plots, we also add a linear regression line to show the trend of the data, the slope of the regression line represents the effect of the weather variable on completion time.

For the **RH** variable, we observed a roughly flat trend for both male and female runners, which means the **RH** level has little effect on completion time. And the gap between the regression lines for different gender groups is relatively small, which means the effect of **RH** level on completion time is similar for different gender. For different age groups, we found the trend stays flat for most of the age groups, except for the 80-99 age group, which has a slightly decreasing trend. This suggests the higher the age, the less effect the **RH** level has on completion time, which is not matching with common sense. This might be due to the small number of participants in the 80-99 age group, causing the result to be deviated.

**Figure 7: RH Effect on Marathon Performance**



For the SR and Wind variable, we observed a similar pattern in RH.

**Figure 8: Wind Effect on Marathon Performance**

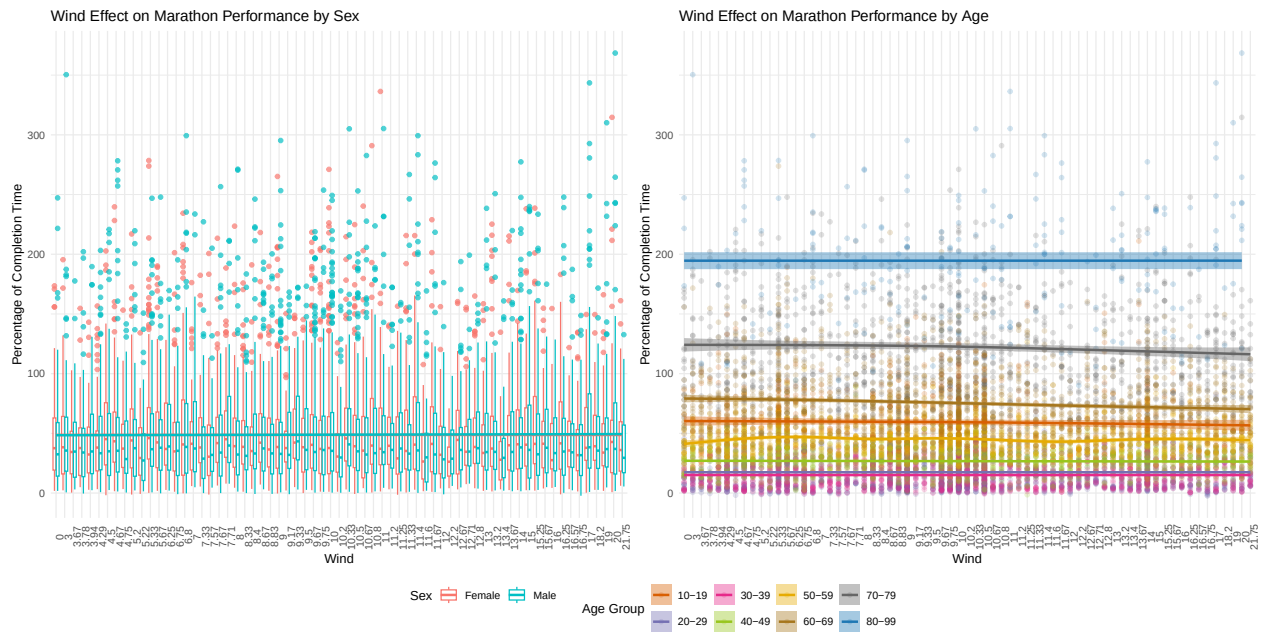
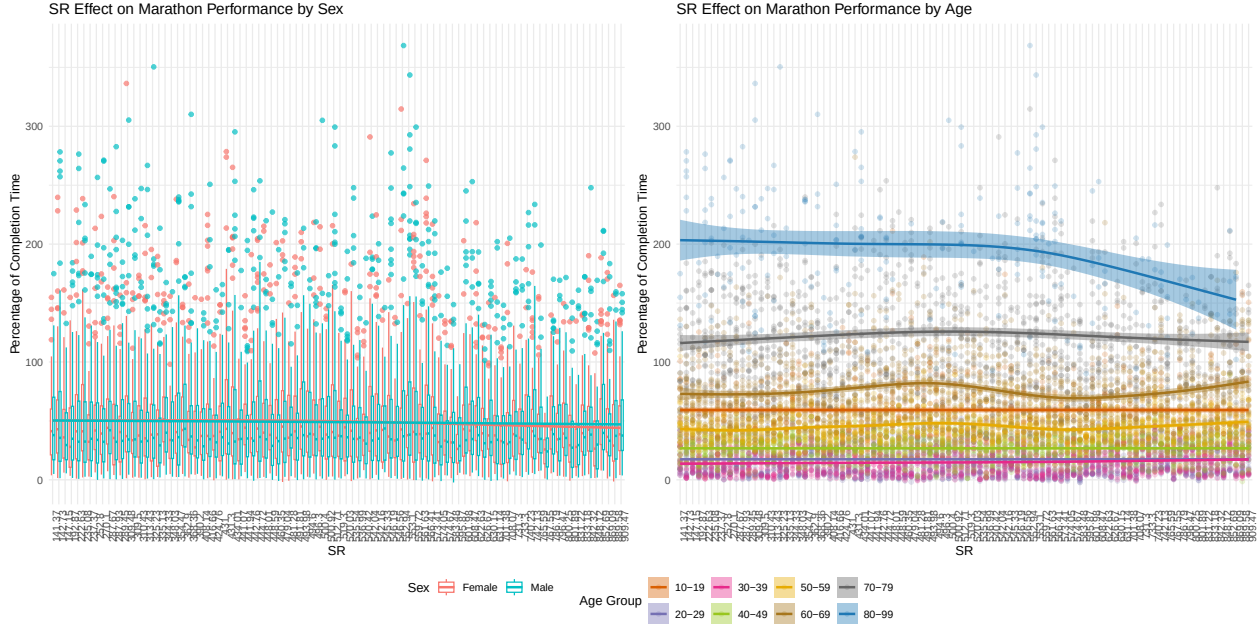


Figure 9: SR Effect on Marathon Performance



## Model Building

The visualization results is a good way to show the general effect or the trend of the data, however, it is not a good way to quantify the effect of the weather variables on performance. In order to quantify the effect, we use linear regression models to reflect the effect. Several models were build.

Firstly, we build a model that includes the **Sex** and **Age** variables. The result shows that the **Sex** variable has a negative effect of -4.7812 on completion percentage, while the **Age** variable has a positive effect of 1.7545 on completion percentage. Both of the P-values are significant. The estimates means that males are expected to have a 4.7812 percent faster completion than female, and the completion percentage will increase by 1.7545 percent for each year increase in the **Age** variable.

Table 4: Sex and Age Model Coefficients

|             | Estimate   | Std. Error | t value    | Pr(> t ) |
|-------------|------------|------------|------------|----------|
| (Intercept) | -30.215186 | 0.8782862  | -34.402438 | 0        |
| SexMale     | -4.781230  | 0.6080130  | -7.863696  | 0        |
| Age         | 1.754557   | 0.0168972  | 103.836895 | 0        |

$$\begin{aligned}
 Model_{sex\_age} : CR_{PERCENTAGE} &= \beta_0 + \beta_1 \times Sex + \beta_2 \times Age \\
 &= -30.2151 - 4.7812 \times Sex_{male} + 1.7545 \times Age
 \end{aligned}$$

The RH model, which includes the **RH**, **Sex**, and **Age** variables as well as the interaction terms between **RH** and **Sex**, and **RH** and **Age**. The result shows that the **RH** variable has a positive effect of 0.1637 on completion percentage. Also, the **Sex** variable has a negative effect of -5.3774 on completion percentage, while the **Age** variable has a positive effect of 1.9105 on completion percentage. The P-values of all of the estimates are significant. The interaction term of **RH** and **Sex** has a positive effect of 0.0128 on completion percentage, while the interaction term of **RH** and **Age** has a negative effect of -0.0036 on completion percentage. The P-value of **RH-Sex** interaction term is 0.4983, which is not significant, suggesting that the **RH** effect does not differ between male and female runners. The estimate of **RH-Age** interaction term is significant, suggesting

that the effect of **RH** on completion percentage is different for different age groups, more specifically, with same level of **RH**, runners will have an extra 0.0036 percent decrease for each year increase in the **Age**.

Table 5: RH Model Coefficients

|                   | Estimate    | Std. Error | t value     | Pr(> t )  |
|-------------------|-------------|------------|-------------|-----------|
| (Intercept)       | -37.1520990 | 1.4553234  | -25.5284152 | 0.0000000 |
| <b>RH</b>         | 0.1637750   | 0.0273895  | 5.9794740   | 0.0000000 |
| <b>SexMale</b>    | -5.3774050  | 1.0078170  | -5.3356960  | 0.0000001 |
| <b>Age</b>        | 1.9105133   | 0.0280386  | 68.1387649  | 0.0000000 |
| <b>RH:SexMale</b> | 0.0128486   | 0.0189750  | 0.6771334   | 0.4983355 |
| <b>RH:Age</b>     | -0.0036852  | 0.0005293  | -6.9624974  | 0.0000000 |

$$\begin{aligned}
Model_{RH} : CR_{PERCENTAGE} &= \beta_0 + \beta_1 \times RH + \beta_2 \times Sex + \beta_3 \times Age + \beta_4 \times RH \times Sex + \beta_5 \times RH \times Age \\
&= \beta_0 + \beta_2 \times Sex + \beta_3 \times Age + (\beta_1 + \beta_4 \times Sex + \beta_5 \times Age) \times RH \\
&= -37.1520 - 5.3774 \times Sex + 1.9105 \times Age \\
&\quad + (0.1637 + 0.0128 \times Sex - 0.0036 \times Age) \times RH
\end{aligned}$$

The **SR** model, which includes the **SR**, **Sex**, and **Age** variables as well as the interaction terms between **SR** and **Sex**, and **SR** and **Age**. The result shows that the **SR** variable has a positive effect of 0.0271 on completion percentage. Also, the **Sex** variable has a negative effect of -6.1501 on completion percentage, while the **Age** variable has a positive effect of 2.0834 on completion percentage. The P-values of all of the estimates are significant. The interaction term of **SR** and **Sex** has a positive effect of 0.0027 on completion percentage, while the interaction term of **SR** and **Age** has a negative effect of -0.0006 on completion percentage. The P-value of **SR-Sex** interaction term is 0.3946, which is not significant, suggesting that the **SR** effect does not differ between male and female runners. The estimate of **SR-Age** interaction term is significant, suggesting that the effect of **SR** on completion percentage is different for different age groups, more specifically, with same level of **SR**, runners will have an extra 0.0006 percent decrease for each year increase in the **Age**.

Table 6: SR Model Coefficients

|                   | Estimate    | Std. Error | t value     | Pr(> t )  |
|-------------------|-------------|------------|-------------|-----------|
| (Intercept)       | -44.0394398 | 2.5607094  | -17.1981401 | 0.0000000 |
| <b>SR</b>         | 0.0271455   | 0.0046829  | 5.7967130   | 0.0000000 |
| <b>SexMale</b>    | -6.1501991  | 1.7682169  | -3.4781928  | 0.0005067 |
| <b>Age</b>        | 2.0834985   | 0.0489854  | 42.5330131  | 0.0000000 |
| <b>SR:SexMale</b> | 0.0027535   | 0.0032347  | 0.8512602   | 0.3946433 |
| <b>SR:Age</b>     | -0.0006503  | 0.0000907  | -7.1669548  | 0.0000000 |

$$\begin{aligned}
Model_{SR} : CR_{PERCENTAGE} &= \beta_0 + \beta_1 \times SR + \beta_2 \times Sex + \beta_3 \times Age + \beta_4 \times SR \times Sex + \beta_5 \times SR \times Age \\
&= \beta_0 + \beta_2 \times Sex + \beta_3 \times Age + (\beta_1 + \beta_4 \times Sex + \beta_5 \times Age) \times SR \\
&= -44.0394 - 6.1501 \times Sex + 2.0834 \times Age \\
&\quad + (0.0271 + 0.0027 \times Sex - 0.0006 \times Age) \times SR
\end{aligned}$$

The **Wind** model, which includes the **Wind**, **Sex**, and **Age** variables as well as the interaction terms between **Wind** and **Sex**, and **Wind** and **Age**. The result shows that the **Wind** variable has a negative effect of -1.0906

on completion percentage. Also, the **Sex** variable has a negative effect of -6.2381 on completion percentage, while the **Age** variable has a positive effect of 1.5773 on completion percentage. The P-values of all of the estimates are significant. The interaction term of **Wind** and **Sex** has a positive effect of 0.1449 on completion percentage, while the interaction term of **Wind** and **Age** has a positive effect of 0.01785 on completion percentage. The P-value of **Wind-Sex** interaction term is 0.3314, which is not significant, suggesting that the **Wind** effect does not differ between male and female runners. The estimate of **Wind-Age** interaction term is significant, suggesting that the effect of **Wind** on completion percentage is different for different age groups, more specifically, with same level of **Wind**, runners will have an extra 0.0178 percent increase for each year increase in the **Age**.

Table 7: Wind Model Coefficients

|              | Estimate    | Std. Error | t value    | Pr(> t )  |
|--------------|-------------|------------|------------|-----------|
| (Intercept)  | -19.4181455 | 2.3291616  | -8.3369678 | 0.0000000 |
| Wind         | -1.0906781  | 0.2176067  | -5.0121527 | 0.0000005 |
| SexMale      | -6.2381655  | 1.6032403  | -3.8909734 | 0.0001004 |
| Age          | 1.5773520   | 0.0447314  | 35.2627148 | 0.0000000 |
| Wind:SexMale | 0.1449589   | 0.1492629  | 0.9711649  | 0.3314874 |
| Wind:Age     | 0.0178579   | 0.0041498  | 4.3033489  | 0.0000170 |

$$\begin{aligned}
Model_{Wind} : CR_{PERCENTAGE} &= \beta_0 + \beta_1 \times Wind + \beta_2 \times Sex + \beta_3 \times Age + \beta_4 \times Wind \times Sex + \beta_5 \times Wind \times Age \\
&= \beta_0 + \beta_2 \times Sex + \beta_3 \times Age + (\beta_1 + \beta_4 \times Sex + \beta_5 \times Age) \times Wind \\
&= -19.4181 - 6.2381 \times Sex + 1.5773 \times Age \\
&\quad + (-1.0906 + 0.14495 \times Sex + 0.0178 \times Age) \times Wind
\end{aligned}$$

The **WBGT** model, which includes the **WBGT**, **Sex**, and **Age** variables as well as the interaction terms between **WBGT** and **Sex**, and **WBGT** and **Age**. The result shows that the **WBGT** variable has a positive effect of 1.2692 on completion percentage. Also, the **Sex** variable has a negative effect of -4.4349 on completion percentage, while the **Age** variable has a positive effect of 1.9852 on completion percentage. The P-values of all of the estimates are significant. The interaction term of **WBGT** and **Sex** has a negative effect of -0.0286 on completion percentage, while the interaction term of **WBGT** and **Age** has a negative effect of -0.0174 on completion percentage. The P-value of **WBGT-Sex** interaction term is 0.7915, which is not significant, suggesting that the **WBGT** effect does not differ between male and female runners. The estimate of **WBGT-Age** interaction term is significant, suggesting that the effect of **WBGT** on completion percentage is different for different age groups, more specifically, with same level of **WBGT**, runners will have an extra 0.0174 percent decrease for each year increase in the **Age**.

Table 8: WBGT Model Coefficients

|              | Estimate    | Std. Error | t value     | Pr(> t )  |
|--------------|-------------|------------|-------------|-----------|
| (Intercept)  | -46.8933940 | 2.2028065  | -21.2880227 | 0.0000000 |
| WBGT         | 1.2692664   | 0.1555470  | 8.1600200   | 0.0000000 |
| SexMale      | -4.4349396  | 1.5240973  | -2.9098796  | 0.0036229 |
| Age          | 1.9852210   | 0.0422178  | 47.0233706  | 0.0000000 |
| WBGT:SexMale | -0.0286229  | 0.1083180  | -0.2642484  | 0.7915934 |
| WBGT:Age     | -0.0174927  | 0.0030175  | -5.7971717  | 0.0000000 |

$$\begin{aligned}
Model_{WBGT} : CR_{PERCENTAGE} &= \beta_0 + \beta_1 \times WBGT + \beta_2 \times Sex + \beta_3 \times Age + \beta_4 \times WBGT \times Sex + \beta_5 \times WBGT \times Age \\
&= \beta_0 + \beta_2 \times Sex + \beta_3 \times Age + (\beta_1 + \beta_4 \times Sex + \beta_5 \times Age) \times WBGT \\
&= -46.8933 - 4.4349 \times Sex + 1.9852 \times Age \\
&\quad + (1.2692 - 0.0286 \times Sex - 0.0174 \times Age) \times WBGT
\end{aligned}$$

From the results above, we can tell that none of the weather variables has different effect among genders, however, they do tend to have different effect among different age groups. In order to reflect the total effect, we combine those models into a general model.

Table 9: General Model Coefficients

|             | Estimate    | Std. Error | t value    | Pr(> t )  |
|-------------|-------------|------------|------------|-----------|
| (Intercept) | -69.4308723 | 4.5839555  | -15.146498 | 0.0000000 |
| RH          | 0.2727244   | 0.0307757  | 8.861691   | 0.0000000 |
| SR          | 0.0434054   | 0.0054814  | 7.918591   | 0.0000000 |
| Wind        | -0.3553057  | 0.2213215  | -1.605383  | 0.1084381 |
| WBGT        | 0.6903328   | 0.1663000  | 4.151129   | 0.0000333 |
| SexMale     | -4.8122171  | 0.6008163  | -8.009464  | 0.0000000 |
| Age         | 2.5651730   | 0.0913176  | 28.090689  | 0.0000000 |
| RH:Age      | -0.0063505  | 0.0006135  | -10.352104 | 0.0000000 |
| SR:Age      | -0.0011180  | 0.0001099  | -10.171410 | 0.0000000 |
| Wind:Age    | 0.0071830   | 0.0043962  | 1.633892   | 0.1023100 |
| WBGT:Age    | -0.0034103  | 0.0033647  | -1.013528  | 0.3108302 |

$$\begin{aligned}
Model_{general_1} : CR_{PERCENTAGE} &= \beta_0 + \beta_1 \times RH + \beta_2 \times SR + \beta_3 \times Wind + \beta_4 \times WBGT \\
&\quad + \beta_5 \times Sex + \beta_6 \times Age \\
&\quad + \beta_7 \times RH \times Age + \beta_8 \times SR \times Age + \beta_9 \times Wind \times Age + \beta_{10} \times WBGT \times Age \\
&= \beta_0 + \beta_5 \times Sex + \beta_6 \times Age \\
&\quad + (\beta_1 + \beta_7 \times Age) \times RH + (\beta_2 + \beta_8 \times Age) \times SR \\
&\quad + (\beta_3 + \beta_9 \times Age) \times Wind + (\beta_4 + \beta_{10} \times Age) \times WBGT \\
&= -69.4308 - 4.8122 \times Sex + 2.5651 \times Age \\
&\quad + (0.2727 - 0.0063 \times Age) \times RH + (0.0434 - 0.0011 \times Age) \times SR \\
&\quad + (-0.3553 + 0.0071 \times Age) \times Wind + (0.6903 - 0.0034 \times Age) \times WBGT
\end{aligned}$$

As we looking at the interaction term, we can easily find out the weather variable that has the greatest impact on runners performance. Considering from the weather condition, when runners have the same age, the WBGT variable has the greatest impact on completion percentage, with a 0.6903 percent increase plus -0.0034 age adjustment for each unit increase in WBGT. Considering from the ageing effect, the Wind variable has the greatest impact on completion percentage, with a 0.0071 age adjustment.

## Discussion

In this report, we analyzed the effect of inner factors (age, sex) and outer factors (weather variables) on marathon performance. Through comprehensive analysis of data from five major marathons, we identified several key findings regarding both physiological and environmental impacts on marathon completion times.

Our analysis revealed that while male and female runners have objectively different race performance with males generally running faster, weather conditions affect both genders similarly, suggesting the performance gap is primarily due to physiological differences rather than differential weather sensitivity. This finding aligns with Ely et al. (2007), who found similar patterns of weather-related performance decrements between male and female elite runners. However, weather conditions do have significantly different effects across age groups. The WBGT variable demonstrated the strongest impact on completion percentage, with a 0.6903 percent increase plus a -0.0034 age adjustment for each unit increase in WBGT. This supports findings from Ely et al. (2010) showing that aerobic performance is degraded in hot environments even with modest hyperthermia.

The relative humidity (RH) and solar radiation (SR) showed comparatively smaller impacts on performance, though their effects were still statistically significant. Interestingly, wind speed demonstrated a unique pattern - while it had a relatively minor direct effect on performance, it showed the strongest interaction with age among all weather variables, suggesting older runners may be particularly sensitive to wind conditions. This age-dependent response to weather conditions likely reflects the reduced thermoregulatory capacity and cardiovascular adaptability documented in older adults.

The progressive impact of increasing WBGT on performance across all age groups supports the concept of a “prescriptive zone” for temperature regulation during exercise, as described by Ely et al. (2007). When environmental conditions push runners outside this zone, performance necessarily declines as the body struggles to maintain thermal homeostasis. This is particularly evident in our finding that the performance impact of adverse weather conditions increases with both age and completion time, suggesting that slower runners face compounded challenges from prolonged exposure to challenging weather conditions.

A limitation of our analysis is that the dataset only captures weather variables without accounting for other important factors such as training status, nutrition, and general health status. Additionally, the smaller sample sizes in the youngest (10-19) and oldest (80-99) age groups may affect the reliability of our findings for these populations. Future research would benefit from incorporating these additional variables and ensuring more balanced representation across age groups.

These findings have important practical implications for runners, race organizers, and medical personnel. They suggest that race organizers should consider developing age-specific guidelines for weather-based race modifications, rather than applying uniform standards across all participants. Runners, particularly older athletes or those expecting longer completion times, should adjust their race strategies and performance expectations based on weather conditions, with special attention to WBGT forecasts. Future research could focus on developing more precise models for predicting individual performance decrements based on both personal factors (age, sex, fitness level) and environmental conditions.

## References

1. Ely, B. R., Cheuvront, S. N., Kenefick, R. W., & Sawka, M. N. (2010). Aerobic performance is degraded, despite modest hyperthermia, in hot environments. *Med Sci Sports Exerc*, 42(1), 135-41.
2. Ely, M. R., Cheuvront, S. N., Roberts, W. O., & Montain, S. J. (2007). Impact of weather on marathon-running performance. *Medicine and science in sports and exercise*, 39(3), 487-493.
3. Kenney, W. L., & Munce, T. A. (2003). Invited review: aging and human temperature regulation. *Journal of applied physiology*, 95(6), 2598-2603.
4. Besson, T., Macchi, R., Rossi, J., Morio, C. Y., Kunimasa, Y., Nicol, C., ... & Millet, G. Y. (2022). Sex differences in endurance running. *Sports medicine*, 52(6), 1235-1257.
5. Yanovich, R., Ketko, I., & Charkoudian, N. (2020). Sex differences in human thermoregulation: relevance for 2020 and beyond. *Physiology*, 35(3), 177-184.

## Code Appendix

```
knitr::opts_chunk$set(echo = FALSE)
knitr::opts_chunk$set(message = FALSE)
knitr::opts_chunk$set(warning = FALSE)
library(mice, warn.conflicts = FALSE)
library(naniar)
library(ggplot2)
library(dplyr)
library(readr)
library(tidyr)
library(readxl)
library(ggpubr)
library(gtsummary)
library(GGally)
library(ggcorrplot)
library(knitr)
library(kableExtra)
library(lubridate)
library(patchwork)
library(introdataviz)

# Load data
marathon_data <- read.csv("../Data/project1.csv")
course_record <- read.csv("../Data/course_record.csv")

# rename the column names that are too long to follow.
colnames(marathon_data)[1] <- "Race"
colnames(marathon_data)[3] <- "Sex"
colnames(marathon_data)[5] <- "Age"
colnames(marathon_data)[6] <- "CR_PERCENTAGE"
colnames(marathon_data)[7] <- "TD"
colnames(marathon_data)[8] <- "TW"
colnames(marathon_data)[9] <- "RH"
colnames(marathon_data)[10] <- "TG"
colnames(marathon_data)[11] <- "SR"

# data type conversion
marathon_data$Year <- as.factor(marathon_data$Year)
marathon_data$Race <- as.factor(marathon_data$Race)
marathon_data$Sex <- as.factor(marathon_data$Sex)
marathon_data$Flag <- as.factor(marathon_data$Flag)

marathon_data$Flag[marathon_data$Flag == ""] <- NA

# replace marathon name with code name in course_record
course_record$Race[course_record$Race == "B"] <- 0
course_record$Race[course_record$Race == "C"] <- 1
course_record$Race[course_record$Race == "NY"] <- 2
course_record$Race[course_record$Race == "TC"] <- 3
course_record$Race[course_record$Race == "D"] <- 4
course_record$Race <- as.factor(course_record$Race)
```

```

# replace gender in course_record
course_record$Gender[course_record$Gender == "M"] <- 1
course_record$Gender[course_record$Gender == "F"] <- 0
course_record$Gender <- as.factor(course_record$Gender)
colnames(course_record)[4] <- "Sex"

# Transform records in course_record into seconds
course_record$CR <- period_to_seconds(hms(course_record$CR))

# Join course_record and marathon_data
marathon_data <- merge(marathon_data, course_record, by = c("Race", "Year", "Sex"))

# calculate the record of each runner
marathon_data$CR <- (1 + marathon_data$CR_PERCENTAGE * 0.01) * marathon_data$CR

marathon_data <- marathon_data %>%
  mutate(Race = case_when(
    Race == 0 ~ "Boston",
    Race == 1 ~ "Chicago",
    Race == 2 ~ "NYC",
    Race == 3 ~ "Twin Cities",
    Race == 4 ~ "Grandma"
  ),
  Sex = case_when(
    Sex == 1 ~ "Male",
    Sex == 0 ~ "Female"
  )) %>%
  mutate(Age_group = cut(Age, breaks = seq(0, 100, by = 10), right = FALSE,
    labels = c("0-9", "10-19", "20-29", "30-39", "40-49",
      "50-59", "60-69", "70-79", "80-89", "90-99")))

marathon_data$Flag <- factor(marathon_data$Flag, levels = c("White", "Green", "Yellow", "Red", "Black"))

# Check for missing values and patterns
miss_plot <- vis_miss(marathon_data) + labs(title = "Figure 1: Missing Data Pattern")
ggsave("../Plots/missing_values_plot.png", plot = miss_plot, width = 15, dpi = 300)
miss_plot

# Check the missing percentage of weather data in each marathon by year
marathon_data %>%
  group_by(Race, Year) %>%
  summarise(missing_percentage = sum(is.na(Flag)) / n()) %>%
  pivot_wider(names_from="Race", values_from = missing_percentage) %>%
  arrange(Year) %>%
  replace_na(list(Boston = 0, Chicago = 0, NYC = 0, `Twin Cities` = 0, Grandmas = 0)) %>%
  kable(caption = "Missing Percentage of Weather Data in Each Race by Year")

# remove missing data
marathon_data <- marathon_data %>% filter(!is.na(Flag))
completion_time_race <- ggplot(marathon_data, aes(x = Year, y = CR)) +
  geom_boxplot(aes(fill = Race), alpha = 0.5) +
  stat_summary(fun = "mean", geom = "point", shape = 23, size = 1, fill = "red") +
  stat_summary(fun = "mean", geom = "line", aes(group = 1), color = "red") +

```

```

facet_wrap(~ Race) +
labs(title = "Figure 2: CR Distribution by Year and Race",
     x = "Year",
     y = "Completion Time (CR)") +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 90, hjust = 1),
  legend.position = "none"
)
ggsave("../Plots/completion_time_race.png", plot = completion_time_race, width = 10, height = 8, dpi = 300)
completion_time_race

tbl_summary(
  marathon_data %>% select(Race, Age_group, Sex),
  by = Race,
  statistic = list(
    Age_group ~ "{n} ({p}%)",
    Sex ~ "{n} ({p}%)",
  )
) %>%
  modify_header(label = "**Age Group**") %>%
  modify_caption("**Number of Participants by Age Group, Sex and Race**")
participants_age_plot <- ggplot(marathon_data, aes(x = Age_group, fill = Age_group)) +
  geom_bar() +
  facet_wrap(~ Race) +
  scale_fill_viridis_d() +
  labs(title = "Number of Participants by Age Group for Each Race",
       x = "Age Group",
       y = "Number of Participants",
       fill = "Age Group") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

ggsave("../Plots/participants_age_plot.png", plot = participants_age_plot, width = 10, height = 8, dpi = 300)
participants_age_plot

marathon_data <- marathon_data %>%
  mutate(Age_group = if_else(Age_group == "90-99", "80-99", Age_group)) %>%
  mutate(Age_group = if_else(Age_group == "80-89", "80-99", Age_group))
sex_distribution_race <- ggplot(marathon_data, aes(x = Sex, fill = Sex)) +
  geom_bar(position = "dodge", alpha = 0.7) +
  facet_wrap(~ Race, scales = "free_y") +
  labs(title = "Sex Distribution by Race",
       x = "Sex",
       y = "Count",
       fill = "Sex") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(face = "italic"),
    legend.position = "none"
  )
ggsave("../Plots/sex_distribution_race.png", plot = sex_distribution_race, width = 10, height = 8, dpi = 300)
sex_distribution_race

```

```

cor_plot <- ggpairs(marathon_data %>% select(TD, TW, RH, TG, SR, DP, Wind, WGBT)) + ggtitle("Figure 3: ")
ggsave("../Plots/cor_plot.png", plot = cor_plot, width = 10, height = 8, dpi = 300)
cor_plot

cr_distribution_plot <- ggplot(marathon_data, aes(x=Race, y = CR_PERCENTAGE, fill=Sex)) +
  geom_split_violin() +
  labs(title = "Figure 4: Distribution of Completion Time Percentage",
       y = "CR Percentage") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1)
  )
ggsave("../Plots/cr_distribution_plot.png", plot = cr_distribution_plot, width = 8, height = 5, dpi = 300)
cr_distribution_plot

age_effect_plot <- ggplot(marathon_data, aes(x = Age, y = CR, color = Sex)) +
  geom_point(alpha = 0.1) +
  geom_smooth(method = "loess", se = TRUE) +
  facet_wrap(~ Race, scales = "free_y") +
  labs(title = "Figure 5: Effect of Age on Marathon Performance by Race",
       x = "Age (yrs)",
       y = "Best Time (CR)",
       color = "Sex") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1),
    legend.position = "top"
  )
ggsave("../Plots/age_effect_plot.png", plot = age_effect_plot, width = 10, height = 8, dpi = 300)
age_effect_plot

# WGBT effects
flag_colors <- c("Green" = "darkgreen",
                 "Yellow" = "orange",
                 "Red" = "darkred",
                 "White" = "darkgrey")

wgbt_effect <- marathon_data %>%
  ggplot(aes(x = Age, y = CR_PERCENTAGE, color = Flag)) +
  facet_wrap(~ Sex, scales = "fixed") +
  geom_point(alpha = 0.1) +
  geom_smooth(aes(group = Flag, color = Flag), se = T, size=0.5) +
  geom_hline(yintercept = 50, linetype = "dashed", color = "blue") +
  geom_hline(yintercept = 100, linetype = "dashed", color = "blue") +
  geom_hline(yintercept = 150, linetype = "dashed", color = "blue") +
  geom_hline(yintercept = 200, linetype = "dashed", color = "blue") +
  scale_color_manual(values = flag_colors) +
  labs(title = "Figure 6: WGBT Effect on Completion Time by Age and Sex",
       x = "Age",
       y = "Percentage of Completion Time",
       color = "Flag") +
  theme_minimal() +

```

```

theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
)
ggsave("../Plots/wgbt_effect.png", plot = wgbt_effect, width = 10, height = 8, dpi = 300)
wgbt_effect

age_group_colors <- c("0-9" = "#1b9e77",
  "10-19" = "#d95f02",
  "20-29" = "#7570b3",
  "30-39" = "#e7298a",
  "40-49" = "#98c61e",
  "50-59" = "#e6ab02",
  "60-69" = "#a6761d",
  "70-79" = "#666666",
  "80-99" = "#1f78b4")

# RH Effects
RH_sex <- marathon_data %>%
  ggplot(aes(x = as.factor(round(RH, 2)), y = CR_PERCENTAGE, color = Sex)) +
  geom_boxplot(alpha = 0.7) +
  geom_smooth(aes(group = Sex), se = FALSE) +
  labs(title = "RH Effect on Marathon Performance by Sex",
    x = "RH",
    y = "Percentage of Completion Time") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90)
  )

RH_age <- marathon_data %>%
  ggplot(aes(x = as.factor(round(RH, 2)), y = CR_PERCENTAGE, fill = Age_group, color = Age_group)) +
  geom_point(alpha = 0.2) +
  geom_smooth(aes(group = Age_group, color = Age_group), se = T) +
  scale_fill_manual(values = age_group_colors) +
  scale_color_manual(values = age_group_colors) +
  labs(title = "RH Effect on Marathon Performance by Age",
    x = "RH",
    y = "Percentage of Completion Time",
    fill = "Age Group",
    color = "Age Group") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90)
  )

# Wind Effects
Wind_sex <- marathon_data %>%
  ggplot(aes(x = as.factor(round(Wind, 2)), y = CR_PERCENTAGE, color = Sex)) +
  geom_boxplot(alpha = 0.7) +
  geom_smooth(aes(group = Sex), se = FALSE) +
  labs(title = "Wind Effect on Marathon Performance by Sex",
    x = "Wind",
    y = "Percentage of Completion Time") +

```

```

    theme_minimal() +
    theme(
      axis.text.x = element_text(angle = 90)
    )

Wind_age <- marathon_data %>%
  ggplot(aes(x = as.factor(round(Wind, 2)), y = CR_PERCENTAGE, fill = Age_group, color = Age_group)) +
  geom_point(alpha = 0.2) +
  geom_smooth(aes(group = Age_group, color = Age_group), se = T) +
  scale_fill_manual(values = age_group_colors) +
  scale_color_manual(values = age_group_colors) +
  labs(title = "Wind Effect on Marathon Performance by Age",
       x = "Wind",
       y = "Percentage of Completion Time",
       fill = "Age Group",
       color = "Age Group") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90)
  )

# SR Effects
SR_sex <- marathon_data %>%
  ggplot(aes(x = as.factor(round(SR, 2)), y = CR_PERCENTAGE, color = Sex)) +
  geom_boxplot(alpha = 0.7) +
  geom_smooth(aes(group = Sex), se = FALSE) +
  labs(title = "SR Effect on Marathon Performance by Sex",
       x = "SR",
       y = "Percentage of Completion Time") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90)
  )

SR_age <- marathon_data %>%
  ggplot(aes(x = as.factor(round(SR, 2)), y = CR_PERCENTAGE, fill = Age_group, color = Age_group)) +
  geom_point(alpha = 0.2) +
  geom_smooth(aes(group = Age_group, color = Age_group), se = T) +
  scale_fill_manual(values = age_group_colors) +
  scale_color_manual(values = age_group_colors) +
  labs(title = "SR Effect on Marathon Performance by Age",
       x = "SR",
       y = "Percentage of Completion Time",
       fill = "Age Group",
       color = "Age Group") +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90)
  )

weather_effect <- (RH_sex | RH_age) / (Wind_sex | Wind_age) / (SR_sex | SR_age) +
  plot_layout(guides = "collect", axis_titles = "collect") &
  theme(legend.position = 'bottom') &

```

```

plot_annotation(
  title = "RH, SR and Wind Effect on Marathon Performance",
  theme = theme(
    plot.title = element_text(size = 15, face = "bold")
  )
)
ggsave("../Plots/weather_effect.png", plot = weather_effect, width = 20, height = 30, dpi = 300)
weather_effect

rh_effect <- (RH_sex | RH_age) +
  plot_layout(guides = "collect", axis_titles = "collect") &
  theme(legend.position = 'bottom') &
  plot_annotation(
    title = "Figure 7: RH Effect on Marathon Performance",
    theme = theme(
      plot.title = element_text(size = 15, face = "bold")
    )
  )
ggsave("../Plots/rh_effect.png", plot = rh_effect, width = 15, height = 8, dpi = 300)
rh_effect

wind_effect <- (Wind_sex | Wind_age) +
  plot_layout(guides = "collect", axis_titles = "collect") &
  theme(legend.position = 'bottom') &
  plot_annotation(
    title = "Figure 8: Wind Effect on Marathon Performance",
    theme = theme(
      plot.title = element_text(size = 15, face = "bold")
    )
  )
ggsave("../Plots/wind_effect.png", plot = wind_effect, width = 15, height = 8, dpi = 300)
wind_effect

sr_effect <- (SR_sex | SR_age) +
  plot_layout(guides = "collect", axis_titles = "collect") &
  theme(legend.position = 'bottom') &
  plot_annotation(
    title = "Figure 9: SR Effect on Marathon Performance",
    theme = theme(
      plot.title = element_text(size = 15, face = "bold")
    )
  )
ggsave("../Plots/sr_effect.png", plot = sr_effect, width = 15, height = 8, dpi = 300)
sr_effect

# sex age model
sex_age_model <- glm(CR_PERCENTAGE ~ Sex + Age, data = marathon_data)
kable(summary(sex_age_model)$coefficients, caption = "Sex and Age Model Coefficients")

# RH model
RH_model <- glm(CR_PERCENTAGE ~ RH + Sex + Age + RH * Sex + RH * Age, data = marathon_data)
kable(summary(RH_model)$coefficients, caption = "RH Model Coefficients")

```

```

# SR model
SR_model <- glm(CR_PERCENTAGE ~ SR + Sex + Age + SR * Sex + SR * Age, data = marathon_data)
kable(summary(SR_model)$coefficients, caption = "SR Model Coefficients")

# Wind model
Wind_model <- glm(CR_PERCENTAGE ~ Wind + Sex + Age + Wind * Sex + Wind * Age, data = marathon_data)
kable(summary(Wind_model)$coefficients, caption = "Wind Model Coefficients")

# WBGT model
WBGT_model <- glm(CR_PERCENTAGE ~ WBGT + Sex + Age + WBGT * Sex + WBGT * Age, data = marathon_data)
kable(summary(WBGT_model)$coefficients, caption = "WBGT Model Coefficients")

# General model with WBGT
general_model <- glm(CR_PERCENTAGE ~ RH + SR + Wind + WBGT + Sex + Age + RH * Age + SR * Age + Wind * Age, data = marathon_data)
kable(summary(general_model)$coefficients, caption = "General Model Coefficients")

```